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Accurate Hybrid Flux Observation Based on Improved Voltage Model for Induction Motor

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ABSTRACT The Hybrid Flux Observation Model (HFOM) is frequently employed in induction motor flux observers to ensure accurate and dynamic responses over the entire speed range. Nevertheless, the voltage model, designed in the stationary reference frame, is susceptible to integral saturation and integral drift issues. These problems lead to diminished estimation accuracy and compromised dynamic performance when conventional HFOM is utilized. To mitigate these challenges, this paper introduces a novel Hybrid Flux Observation Model (NHFOM), founded on a voltage model in the synchronous rotating reference frame. Furthermore, this paper explores compensation strategies for phase deviation during model transition and proposes a delay time compensation strategy to mitigate the impact of control system delay on flux observation accuracy at high speeds. Experimental results affirm that the proposed NHFOM enhances the accuracy of flux observation and the dynamic response performance of the motor.

INDEX TERMS Induction motor, Voltage model, Integral saturation, Integral drift, Hybrid flux observation model.

I. INTRODUCTION

The induction motor, characterized by robustness, high reliability and strong adaptability to environmental conditions, is extensively utilized in electric vehicles, port transport equipment, mine electric transmission equipment, and more [1, 2]. Such applications necessitate the induction motor to exhibit high-performance speed control characteristics, including rapid torque response, a wide speed control range, and excellent dynamic performance in the field-weakening region. To achieve high-performance drive in induction motors, control strategies such as direct torque control or vector control are typically implemented [3, 4]. Both strategies require accurate flux observation, which directly influences the torque response characteristics and speed regulation performance of the induction motor [5, 6].

In order to enhance the accuracy of flux observation, advanced techniques like the extended Kalman filter [7], Sliding Mode Observer [8], and the robust adaptive state observer [9] have been investigated. However, these methods necessitate complex observer structures, high computational requirements, and dependency on high-performance digital control systems. To reduce complexity, full-order or reduced-

order flux observers have been proposed [10, 11]. However, the feedback matrix of these observers demonstrates different optimized configurations under varying operating conditions, complicating observer design across the entire speed range [12].

In the realm of induction motor flux observation, the current model and voltage model, based on the motor's model equation, are prevalently utilized to observe the motor's flux [13]. The voltage model performs well due to its robustness to the mismatch of rotor parameters such as R_r and L_m . Yet, due to the offset component in the output voltage of the pulse-width modulated inverter, the voltage model's integral calculation faces issues of integral saturation [14]. Further, the initial value problem of pure integral leads to integral drift problems, making the voltage model challenging to design and implement [15, 16].

Various approaches based on the low-pass filter (LPF) have been designed to rectify the integral saturation and integral drift problems. However, each of these solutions comes with its own set of challenges and limitations. LPF has been employed in [17] in place of the pure integral, however the flux's amplitude and phase are found distorted. A

programmable cascade of low-pass filters has been suggested in [18] to reset a pure integrator and get rid of offset problems, though at low frequencies researchers have found instability and a flux estimation error. To improve performance in low- and zero-speed conditions, the combination structure of a first-order low-pass filter and a first-order high-pass filter has been introduced in [19]. However, the motor's ability to regulate speed is impacted by the filter's cut-off frequency. The closed-loop type flux estimator has been presented in [20, 21] to address the voltage model's two primary drawbacks and to improve stability and accuracy at low frequencies. However, this method requires a trade-off between the distortion level and estimator response speed.

An alternative approach to address the issues of integral saturation and integral drift in the voltage model has been proposed in [22-26]. This approach suggests combining the voltage- and current-model based observers into a hybrid flux observer to overcome the problems of integral saturation and integral initial value. The hybrid flux observer provides a smooth transition between the voltage model and current model near a crossover frequency. The "Observer characteristic function method" is proposed in [22] to compensate for the phase deviation generated during the transition of voltage and current models. A Gopinath-style discrete-time hybrid flux observer is developed in [23] to estimate flux linkage needed for accurate deadbeat-direct torque and flux controller implementation. The small-signal model of the hybrid flux observer is studied in [24] to perform stability analysis. However, the hybrid flux observer has a critical weakness in parametric sensitivity in the mid-speed region due to the effect of the angle of its characteristic function. [14] discusses the analysis of parameter sensitivity of the hybrid flux observer and presents a method for a voltage model based on the Gopinath model. In [25], a distortion-minimizing hybrid flux observer is proposed, which is implemented with voltage- and current-model based flux observers and harmonic extractors to suppress the effects of harmonics. In [26], flux weakening control and parameter sensitivity with the Gopinath-style flux observer are studied to achieve the maximum torque per flux operation. These approaches effectively resolve the problems of integral saturation and integral drift in the voltage model while enhancing the robustness of the model against parameter variations. However, the use of a pure integral voltage model based on the stationary reference frame in these methods introduces challenges related to integral saturation and integral drift. To mitigate these issues, a PI controller is employed to incorporate the flux difference between the current model and the voltage model into the input of the voltage model integrator. While this approach suppresses integral saturation and integral drift, it comes at the expense of reduced integral accuracy in the voltage model. Additionally, the pure integral voltage model in the stationary reference frame is susceptible to the delay of the control system in the field-weakening region, further deteriorating the accuracy. Consequently, the

observation accuracy of the traditional hybrid flux observation model is compromised.

To address the issue of integral saturation and integral drift of the voltage model, this paper explores a voltage model with flux orientation angle feedback closed-loop in the synchronous rotating reference frame. We designed a new hybrid flux observation model (NHFOM) based on the improved voltage model to enhance accuracy and dynamic performance. Furthermore, we modify the characteristic function of the NHFOM to compensate for the phase deviation during model transition. To further improve the accuracy of flux observation across the entire speed range, we proposed a delay compensation strategy based on the NHFOM. Experimental results revealed that the proposed NHFOM can accurately observe the flux over a wide speed range, thereby enabling high-performance control of the induction motor. This work contributes to the literature by providing an innovative solution to the integral saturation and integral drift problems that have traditionally hindered the effectiveness of voltage models in the context of induction motors.

II. NHFOM BASED ON IMPROVED VOLTAGE MODEL

This paper proposes a voltage model rooted in the synchronous rotating reference frame. This model aims at the issues of integral saturation and integral drift via the feedback loop of flux orientation angle. By combining the current model and the proposed voltage model, a new hybrid flux observation model (NHFOM) is also proposed.

A. IMPROVED VOLTAGE MODEL

In the synchronous rotating reference frame oriented by rotor flux, $\psi_{rq}^v = 0$. The complex form of rotor induced electromotive force can be expressed as follows:

$$e_r = \frac{d\psi_r^v}{dt} = \frac{d\psi_{rd}^v}{dt} e^{j\theta} + j\omega_e \psi_{rd}^v e^{j\theta} \quad (1)$$

The dq components are given as:

$$\begin{cases} e_{rd} = \frac{d\psi_{rd}^v}{dt} \\ e_{rq} = \omega_e \psi_{rd}^v \end{cases} \quad (2)$$

Then, the rotor flux and its rotational angular velocity are as follows:

$$\begin{cases} \psi_{rd}^v = \int e_{rd} dt \\ \omega_e = \frac{e_{rq}}{\psi_{rd}^v} \end{cases} \quad (3)$$

The rotor flux orientation angle in voltage mode is:

$$\theta = \int \omega_e dt \quad (4)$$

In these equations, e_r , e_{rd} and e_{rq} denote the complex vector and d-q components of the induced electromotive force of the rotor respectively. ψ_r^v , ψ_{rd}^v and ψ_{rq}^v represent the complex vector and the d-q axis components of rotor flux respectively. ω_e represents the synchronous angular velocity

of the rotor flux, and θ represents the rotor flux orientation angle of the voltage model.

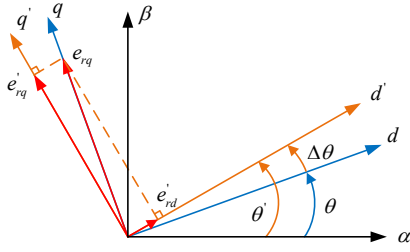


FIGURE 1. Actual and estimated synchronous rotating reference frame.

The proposed voltage model resolves the issues of integral saturation and integral drift through the implementation of a closed-loop flux orientation angle. As demonstrated in Fig. 1, the actual rotor flux orientation resides on the d-axis, whereas the rotor flux orientation observed by the voltage model is on the d'-axis. When ψ_r^v is under steady-state conditions, $e_{rd} = 0$. If integral drift advances the observed flux angle by $\Delta\theta$ relative to the actual flux angle, then the component relationships of the rotor electromotive force in the d-q and d'-q' reference frame are such that $e_{rd}' > e_{rd}$, $e_{rq}' < e_{rq}$. According to equation (3), as $\psi_{rd}^v > \psi_{rd}^v$ and $\omega_e' < \omega_e$, $\Delta\theta$ gradually decreases, bringing the d' axis closer to the d axis until $\Delta\theta = 0$, where finally $\theta' = \theta$. This method effectively suppresses integral drift and saturation.

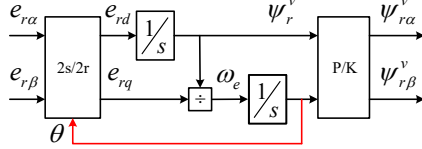


FIGURE 2. Flux observation block diagram of the voltage model.

Fig. 2 illustrates the block diagram of the proposed voltage model, where ψ_{ra}^v and ψ_{rb}^v are the components of the observed flux of the voltage model in the stationary reference frame. Rotor flux can be accurately observed when the motor operates in the mid-high speed region. However, at the low-speed region, due to the increase in the resistance voltage drop of the stator, the observation error of the rotor induced electromotive force rises. Therefore, the voltage model cannot be utilized to observe the flux at low speeds.

B. CURRENT MODEL

In the synchronous rotating reference frame oriented by rotor flux, $\psi_{rq}^v = 0$, the expressions for the rotor flux and its synchronous angular speed are as follows:

$$\psi_{rd}^i = \frac{L_m}{T_r P + 1} i_{sd} \quad (5)$$

$$\begin{cases} \omega_s = \frac{L_m i_{sq}}{T_r \psi_{rd}^i} \\ \omega_1 = \omega_r + \omega_s \end{cases} \quad (6)$$

The rotor flux orientation angle in current mode is:

$$\varphi = \int \omega_1 dt \quad (7)$$

In these formulas, T_r and L_m denote the rotor time constant and mutual inductance, i_{sd} and i_{sq} denote the dq components of the stator current. ψ_{rd}^i represents the d-axis component of the rotor flux, and ω_s and ω_1 stand for the slip angular velocity and synchronous angular speed, respectively.

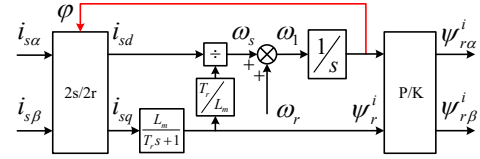


FIGURE 3. Flux observation block diagram of the current model.

Fig. 3 presents the block diagram of the current model based on the synchronous rotating reference frame, where ψ_{ra}^i and ψ_{rb}^i are the components of the flux observed by the current model in the stationary reference frame. The feedback of flux orientation angle realizes the closed-loop observation of flux, thus enhancing the stability of the current model. However, the accuracy of flux estimation by the current model relies on precise modeling of motor parameters. Rotor parameters fluctuate with the increase of rotational speed, and temperature variations can also cause changes in rotor parameters, thereby reducing the accuracy of flux estimation. Consequently, in the middle and high-speed regions, the error of flux observation by the current model may significantly increase.

C. SMOOTH TRANSITION BETWEEN MODELS

In order to exploit the benefits of both the voltage and current models, a hybrid approach is suggested. This approach entails using the current model during low-speed operation and the voltage model during middle and high-speed operation. This combination effectively improves the accuracy and dynamic performance of flux observation across all speed ranges. To facilitate this, the NHFOM, as demonstrated in Fig. 4, is explored in this study.

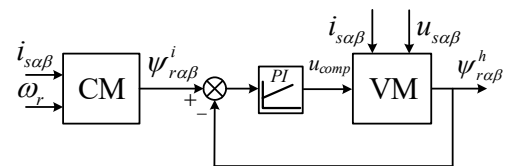


FIGURE 4. Flux observation block diagram of the proposed NHFOM.

The NHFOM is conceptualized as a linear system with two inputs (i.e., the fluxes of the voltage model and current model) and one output (i.e., the observed flux). A PI controller is employed to automate the transition between models. The expression of the estimated rotor flux is as follows:

$$\psi_r^h = \psi_r^v G_v(s) + \psi_r^i G_i(s) \quad (8)$$

In this equation,

$$\begin{cases} G_v(s) = \frac{s^2}{s^2 + k_p s + K_i} \\ G_i(s) = \frac{K_p s + K_i}{s^2 + k_p s + K_i} \\ G_v(s) + G_i(s) = 1 \end{cases}$$

where ψ_r^v represents the estimated rotor flux from the voltage model and ψ_r^i represents the estimated rotor flux from the current model. $G_i(s)$ is a low-pass filter and $G_v(s)$ is a high-pass filter. By judiciously allocating the proportional integral coefficient K_p and K_i , the frequency characteristics of the NHFOM can be adjusted and the automatic transition between the current and voltage models can be achieved.

The selection principle of K_p and K_i is such that the current model dominates at low frequencies and the voltage model prevails at middle and high frequencies. If λ_1 and λ_2 are the two poles of $G_i(s)$ and $G_v(s)$, then $K_p = \lambda_1 + \lambda_2$ and $K_i = \lambda_1 \lambda_2$ hold true, and typically $\lambda_1 \in [2, 5]$ rad/s, $\lambda_2 \in [20, 30]$ rad/s. When $\lambda_1 = 3$ rad/s, $\lambda_2 = 25$ rad/s, the Bode diagram of $G_i(s)$ and $G_v(s)$ is shown in Fig. 5, and the frequency of the transition point in the figure is approximately 4.4 Hz.

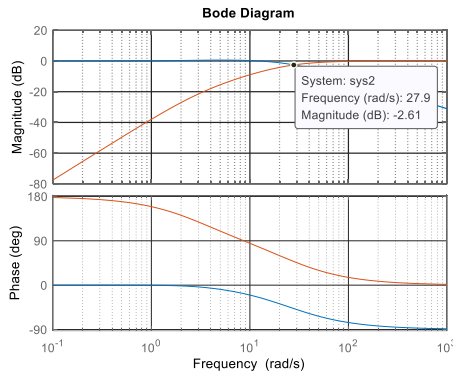


FIGURE 5. Bode diagram for $G_i(s)$, $G_v(s)$

Upon composing the voltage and current models with the PI controller, the NHFOM accrues the advantages of both models in their respective speed regions, and facilitates a smooth transition between the two. This NHFOM, based on the proposed voltage and current models, exhibits stronger parameter robustness, greater accuracy in observation, and a wider range of speed adaptability.

III. COMPENSATION FOR THE PROPOSED NHFOM

A. PHASE COMPENTION

As deduced from the prior analysis, the NHFOM's output in the motor's low-speed region relies predominantly on the flux estimated by the current model. Conversely, in the high-speed region, the NHFOM's output is majorly dependent on the flux

estimated by the voltage model. Crucially, during the intermediate speed section, where a transition occurs from the current model to the voltage model, it is desirable for the output flux of the NHFOM to retain phase consistency throughout the transition process of the two models.

As per equation (8), the following equation holds:

$$\psi_s^h = \psi_s^i [1 - H(s)] + \psi_s^v H(s) \quad (9)$$

$$\text{where, } H(s) = \frac{s^2}{s^2 + k_p s + K_i}$$

In a steady-state scenario, the frequency-response function of $H(s)$ is as follows, which is also illustrated in Fig. 6:

$$H(j\omega_e) = \frac{\omega_e^2}{\sqrt{(K_p \omega_e)^2 + (K_i - \omega_e^2)^2}} e^{j\alpha} \quad (10)$$

where, $\alpha = \pi - \tan^{-1} \left[\frac{k_p \omega_e}{k_i - \omega_e^2} \right]$ and ω_e denotes the electrical angular velocity of the rotor flux.

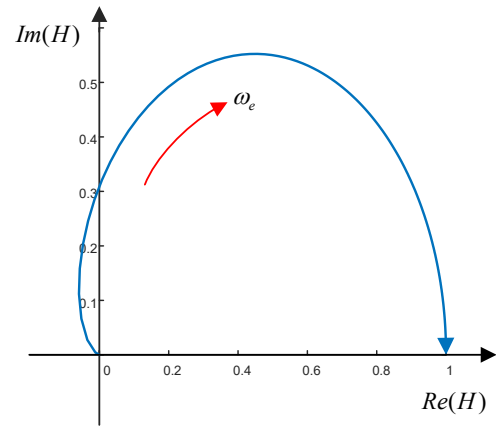


FIGURE 6. Frequency response function of the observer characteristic function, as given in equation (10).

From equation (10), it is evident that the phase angle of the frequency characteristic function $H(s)$ will alter with the rotor's rotational speed. The presence of $e^{j\alpha}$ can potentially induce phase deviation in the observed flux during the transition from the current model to the voltage model, as exemplified in Fig. 6. Further, parameter error and measurement error can exacerbate the accuracy of flux estimation. Consequently, to rectify the flux observation deviation instigated by phase deviation, phase compensation is necessitated.

By introducing a compensation factor, $e^{-j\alpha}$, the resultant expression of $H(s)$ after phase compensation is as follows:

$$H'(s) = H(s) e^{-j\alpha} \quad (11)$$

The frequency-response function of $H'(s)$ can be described as follows, also depicted in Fig. 7:

$$H'(j\omega_e) = \frac{\omega_e^2}{\sqrt{(K_p \omega_e)^2 + (K_i - \omega_e^2)^2}} e^{j0} \quad (12)$$

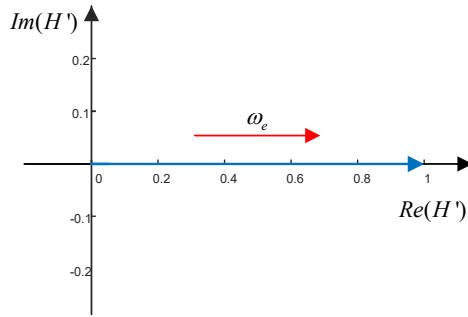


FIGURE 7. Frequency response function of the observer characteristic function, as expressed in equation (12).

Fig. 7 demonstrates that the phase angle of $H'(j\omega_e)$ is zero, unaltered by the rotational speed. Therefore, during the transition from the current model to the voltage model, there will be no phase deviation in the observed flux of the hybrid model. Consequently, the estimated flux will transition from the current model to the voltage model following the optimal trajectory.

Upon the introduction of phase compensation, the NHFOM expression in the stationary reference frame becomes:

$$\lambda_{ra\beta}^h = \psi_{ra\beta}^h e^{-j\alpha} + \psi_{ra\beta}^i (1 - e^{-j\alpha}) \quad (13)$$

Fig. 8 presents the block diagram of the NHFOM, incorporating phase compensation (PC).

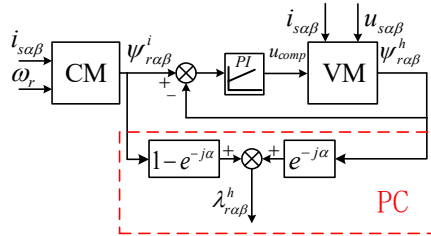


FIGURE 8. NHFOM with phase compensation.

The above analysis substantiates that by utilizing the NHFOM with phase compensation, the phase deviation during the transition process from the current model to the voltage model can be eliminated, thereby enhancing the accuracy of rotor flux observation.

B. DELAY TIME COMPENSATION

In the high-speed field-weakening region, the decline of carrier ratio results in delay time that diminishes the accuracy of flux observation, thereby critically impairing the control performance of the motor. To rectify this, the control system's delay time must be compensated.

The time delay in flux observation within a digital control system encompasses three facets: sampling delay DT_{sample} , control calculation delay $DT_{control}$ and PWM output delay DT_{pwm} . Applying the volt-second balance principle, DT_{pwm} is half of the converter's switching period. $DT_{control}$ is the time taken by the digital processor to execute a motor control algorithm. DT_{sample} is the sampling time of the voltage and current signal. As the sampling delay is relatively minor, this

paper considers only $DT_{control}$ and DT_{pwm} . The digital control system delay time T_d is expressed by:

$$T_d \approx DT_{pwm} + DT_{control} \quad (14)$$

The transfer function correlating the observed flux to the actual flux of the motor is given by:

$$G_\psi = \frac{\psi_{est}(s)}{\psi_{real}(s)} = e^{-T_d s} \quad (15)$$

where, $\psi_{est}(s)$ represents the observed value of the flux, while $\psi_{real}(s)$ represents the actual value of the flux.

Due to the existence of the electromechanical system's moment of inertia, the angular velocity of the flux remains unchanged within the delay time T_d . The relation between the angle of flux delay θ_d and T_d is expressed as:

$$\theta_d = \omega_e T_d \quad (16)$$

To compensate for the delay time T_d , the flux delay angle θ_d can be compensated. Thus, the compensated flux orientation angle is as follows:

$$\theta_\lambda' = \theta_\lambda + \theta_d \quad (17)$$

In this equation, θ_λ represents the rotor flux angle observed by the NHFOM, while θ_λ' represents the rotor flux angle after incorporating delay time compensation.

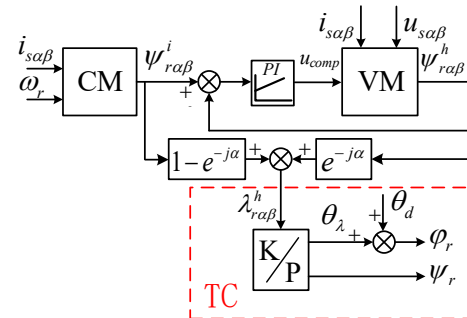


FIGURE 9. NHFOM with delay time compensation.

Fig. 9 illustrates the NHFOM with delay time compensation (TC). By implementing delay time compensation, the issue of reduced accuracy in flux observation caused by the digital control system's delay is resolved. This improves the stability of the current loop and enhances the dynamic response performance of torque when the motor operates in the high-speed region.

By adopting both phase compensation and delay time compensation strategies, the NHFOM's observation accuracy across the full velocity region is enhanced. This results in an improved control system stability and optimized dynamic response performance of the torque under field-weakening conditions.

IV. EXPERIMENTAL RESULTS

The performance of the proposed NHFOM was examined through vector control experiments of an induction motor, conducted on the experimental platform depicted in Fig. 10. The experimental platform includes a core board, a three-level

converter, an oscilloscope, and an induction motor (Motor 1). The core board is equipped with the TMS320F28377D digital control chip from Texas Instruments, in tandem with a digital-to-analog conversion circuit, enabling the observation of relevant digital quantities in the control algorithm. The three-level converter consists of a three-level rectifier and a three-level inverter. Table 1 provides the main parameters of the induction motor utilized in the experiment, with Fig. 11 presenting the block diagram of the vector control system.

TABLE I
PARAMETERS OF A INDUCTION MOTOR

Parameter	Value
Rated voltage	660 V
Rated current	410 A
Stator resistance	0.0796 Ω
Rotor resistance	0.0338 Ω
Stator inductance	0.0223 H
Rotor inductance	0.0223 H
Mutual inductance	0.0217 H

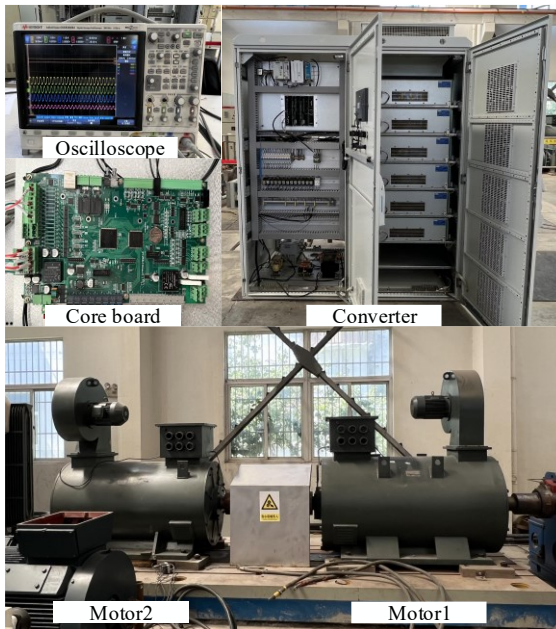


FIGURE 10. Experimental Platform

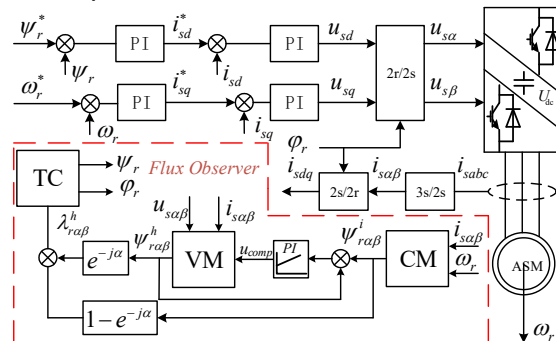


FIGURE 11. Control block diagram of the vector control system.

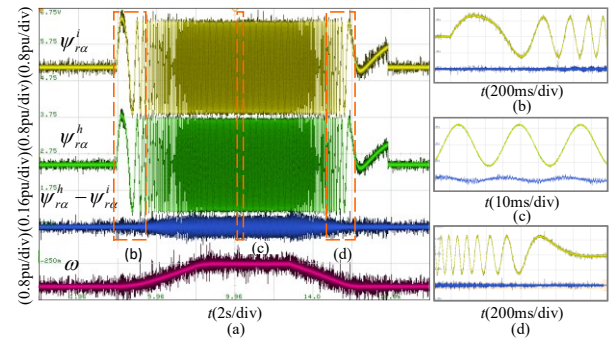


FIGURE 12. Flux components of NHFOM during an acceleration-deceleration cycle. (a) Overall diagram (b) Enlarged view of start-up to acceleration section (c) Enlarged view of constant speed section (d) Enlarged view of deceleration to parking section.

Fig. 12 depicts the results of rotor flux observation by NHFOM during an acceleration-deceleration cycle. In Fig. 12(a), the divergence of flux as observed by the voltage model and the current model is demonstrated, indicating an increase in flux deviation with the rotor speed. The maximum ripple of rotor flux deviation throughout the entire operational cycle, as shown in Fig. 12(c) at the rated speed, is merely $\pm 3.1\%$.

The rotor flux can still be accurately observed at zero-speed, as validated by Fig. 12(b) and (d). The consistency of the flux components between the voltage model and the current model, as shown in Fig. 12(b)-(d), underscores that the observed flux aligns well during the dynamic and steady-state processes of acceleration and deceleration. Thus, it is evident that NHFOM offers beneficial observation results throughout all stages of motor operation.

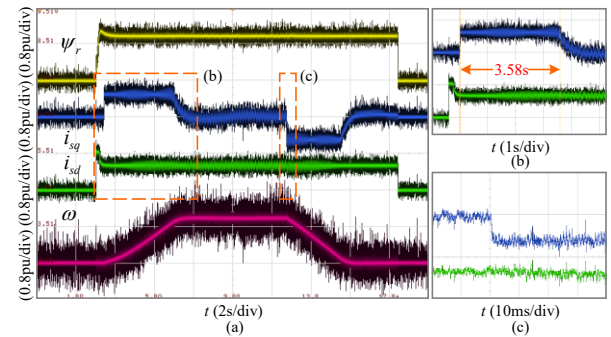


FIGURE 13. Experimental Results of the HFOM proposed in [14] subjected to a step speed change within the rated speed range. (a) Overall diagram (b) Enlarged view of acceleration section (c) Enlarged view on deceleration transients.

Fig. 13 and 14 provide a comparative analysis of the HFOM proposed in [14] and the NHFOM, subjected to a step speed change within the rated speed range. Both models were not implemented with phase compensation and delay time compensation. As demonstrated in Fig. 13(a) and 14(a), the HFOM and the uncompensated NHFOM both display commendable vector control performance.

Upon comparing Fig. 13(b) and 14(b), it is evident that the uncompensated NHFOM offers a reduced acceleration time relative to the HFOM. Furthermore, the comparison of Fig. 13(c) and 14(c) reveals that the uncompensated NHFOM provides a faster torque response and superior dynamic

performance than the HFOM. This affirms the enhanced torque response performance of the uncompensated NHFOM in vector control.

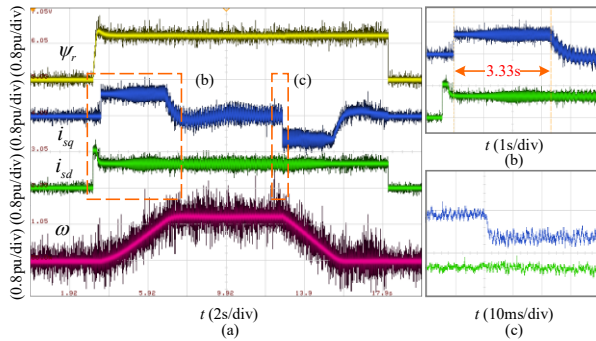


FIGURE 14. Experimental Results of the uncompensated NHFOM subjected to a step speed change within the rated speed range. (a) Overall diagram (b) Enlarged view of acceleration section (c) Enlarged view on deceleration transients.

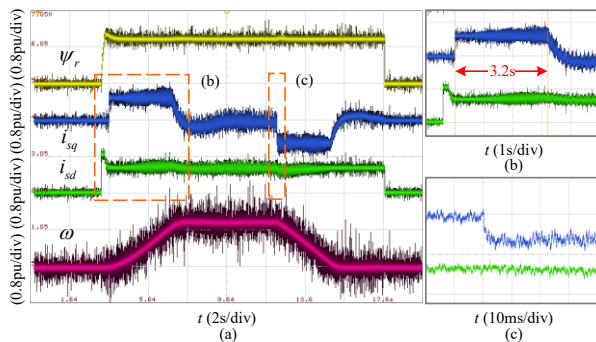


FIGURE 15. Experimental Results of the compensated NHFOM subjected to a step speed change within the rated speed range. (a) Overall diagram (b) Enlarged view of acceleration section (c) Enlarged view on deceleration transients.

The results of Fig. 14 and 15 offer a comparison of the acceleration and deceleration performance of the motor before and after NHFOM compensation, within the rated speed range. As shown in Fig. 14(a) and 15(a), the rotor flux amplitude remains largely stable during the dynamic acceleration and deceleration processes. When a torque step change occurs, the excitation current exhibits excellent decoupling tracking performance, denoting a high-quality dynamic response in the vector control system.

Comparisons between Fig. 14(b) and 15(b) suggest that the compensated NHFOM achieves superior acceleration performance and a shorter acceleration time under vector control. Similarly, Fig. 14(c) and 15(c) comparisons reveal a quicker torque dynamic response with NHFOM compensation in vector control. These experimental results affirm that the application of a compensation strategy within NHFOM enhances flux observation accuracy and improves the dynamic response performance of vector control.

Fig. 16 and 17 display a comparison of experimental results for NHFOM in the transition from zero-speed to 1.8 times the rated speed, both before and after compensation. As shown in Fig. 16(a) and 17(a), the implementation of a compensation strategy enables VMHFM to achieve superior acceleration performance and a shorter acceleration time during the

transition to the field-weakening condition. In the step deceleration from the field-weakening condition, the compensated VMHFM demonstrates a faster adjustment of the magnetic flux amplitude tracking.

Comparing Fig. 16(b)-(c) and 17(b)-(c), it becomes apparent that the NHFOM maintains satisfactory steady-state performance in the field-weakening region, both pre- and post-compensation. With the inclusion of the compensation strategy, the torque dynamic response in the field-weakening region improves, indicating enhanced flux observation accuracy in the high-speed region. These experimental results underscore that the compensated NHFOM improves flux observation accuracy and optimizes the dynamic performance of the induction motor in the field-weakening region.

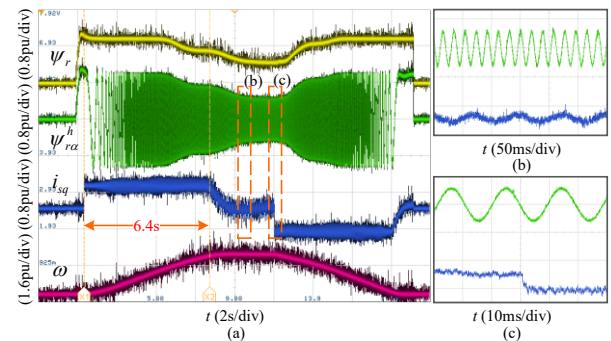


FIGURE 16. Experimental results of the uncompensated NHFOM subjected to a step speed change within the field-weaking range. (a) Overall diagram (b) Enlarged view of constant speed section (c) Enlarged view on deceleration transients.

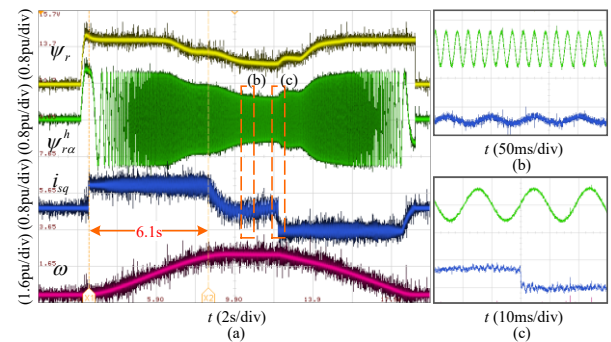


FIGURE 17. Experimental results of the compensated NHFOM subjected to a step speed change within the field-weaking range. (a) Overall diagram (b) Enlarged view of constant speed section (c) Enlarged view on deceleration transients.

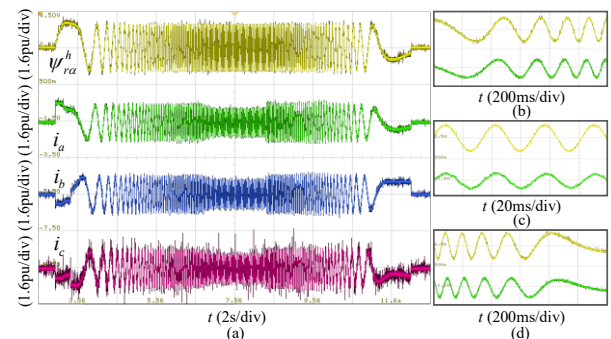


FIGURE 18. Flux and current of motor during an acceleration-deceleration cycle. (a) Overall diagram (b) Enlarged view of acceleration section (c) Enlarged view of constant speed section (d) Enlarged view of deceleration section.

In Fig. 18, the compensated NHFOM demonstrates the observed flux and actual motor current during an acceleration-deceleration cycle. Comparing Fig. 18(b)-(d), it is evident that the motor control system employing the compensated NHFOM generates output current with a desirable sinusoidal waveform and low harmonic distortion.

The experimental results validate that the proposed NHFOM exhibits exceptional performance in flux observation. By implementing a compensation strategy, the accuracy of flux observation is further improved, enabling the induction motor to exhibit rapid torque response, a wide speed control range, and outstanding dynamic performance, even in the field-weakening region. This strategy effectively realizes a wide speed range and high-performance speed control for induction motors.

V. CONCLUSION

This paper proposed a NHFOM based on an improved voltage model for accurate flux observation in the wide speed region of induction motors. Firstly, a voltage model with a closed-loop flux orientation angle feedback in the synchronous rotating reference frame was investigated. Based on this improved voltage model, the new hybrid flux observation model was designed. Secondly, the mechanism of phase deviation in the middle speed region of the NHFOM was examined, leading to the proposal of a phase compensation strategy. Thirdly, the influence of the control system delay on the NHFOM was explored, resulting in the proposal of a delay time compensation strategy. Finally, the proposed NHFOM was compared with the conventional HFOM, which verified the effectiveness and superiority of the proposed algorithm. Experimental results revealed that the NHFOM can accurately observe the flux over a wide speed range, thereby enabling high-performance control of the induction motor.

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